Agenthon 2026

T2 Quant V5

Joint Dependence, Copula Shrinkage + Generalization Report

Core question	Can PCA2 keep its strong marginals while a separate dependence layer fixes the remaining joint variogram weakness?
V5 research strategy	Change only the joint rank/copula structure whenever possible, so marginal and tail scores remain mechanically unchanged.
Final V5 outcome	Full Reference copula, fixed 25% shrinkage, and a past-performance adaptive gate all failed to generalize. The final decision was to keep PCA2 native dependence for V6.

What this report should make clear

- Why V5.0 used rank reordering as a clean dependence-only diagnostic.
- Why the first election-unit result looked unusually strong.
- How six-unit cross-validation overturned the single-unit conclusion.
- Why 25% shrinkage looked promising in development but failed on fresh holdout units.
- Why a chronological adaptive selector also failed to generalize.
- What negative V5 results imply for the final V6 architecture.

Core V5 principle	A dependence model should not be accepted because it improves one joint task. It must survive asset-class variation, multiple RNG seeds, fresh units, and chronological holdout testing.

All results are local pseudo-out-of-sample research experiments, not official sealed leaderboard scores.

1. Starting point – Joint remained the bottleneck

By the end of V4, the PCA2 backbone and the text-tail overlay were relatively stable. The unresolved question was whether cross-asset and cross-horizon dependence could be improved without sacrificing those marginals.

PCA2 strength	Good marginal and tail calibration relative to the Reference in the original UST study.
Persistent weakness	Joint variogram remained unstable across model variants.
V5 experimental rule	Whenever possible, preserve every one-dimensional sample value and change only how dimensions are paired inside a draw.

Keep marginals fixed -〉 change copula/rank structure only -〉 re-score Joint and Composite

Why this matters	If marginal and tail scores remain exactly unchanged, any score movement can be attributed directly to the dependence intervention. This made V5.0 a cleaner diagnostic than another full model rewrite.

2. V5.0 – reference-copula rank reordering

For each dimension, V5.0 sorted the PCA2 marginal draws but assigned those sorted values according to the rank ordering of a Reference Gaussian sample. This is an Iman-Conover-style rank-reordering diagnostic.

Component	V5.0 treatment
Marginal values	Exactly the same PCA2 values before and after reordering
Tail values	Exactly the same because each marginal sample is preserved
Joint structure	Ranks across dimensions follow the Reference Gaussian driver
Diagnostic goal	Measure how much joint score changes when only dependence is altered

PCA2 marginals + Reference rank copula

Sanity check The maximum marginal-preservation error was exactly zero in the first diagnostic run, and the tail delta was also zero by construction.

3. Initial development result – a very strong signal

On t2-F3-election-2024-joint, the dependence-only intervention looked highly successful. Five 5,000-draw RNG seeds were then used to test whether the result was Monte Carlo noise.

Seed	PCA2 Composite	Ref-copula Composite	Reference Composite
101	1.9992	1.9619	1.9621
202	2.0064	1.9585	1.9604
303	2.0036	1.9560	1.9579
404	2.0079	1.9596	1.9621
505	1.9981	1.9515	1.9545

Mean Composite delta vs PCA2	-0.045531
Mean Joint delta vs PCA2	-0.151771
Seeds improving Composite vs PCA2	5 / 5
Seeds improving Joint vs PCA2	5 / 5
Mean Composite delta vs Reference	-0.001898

Development conclusion	The bottleneck diagnosis was real for this unit: changing only dependence materially improved Joint and Composite while leaving Marginal and Tail untouched. The risk was whether this copula structure generalized.

4. Six-unit cross-validation – V5.0 did not generalize

A deliberately diverse validation panel was used: high-dimensional FX, Treasury curves, mixed rates/FX, and factor returns. The original election development unit was excluded.

Validation unit	Composite % vs PCA2	Joint % vs PCA2
Term-premium steepener 2023	-2.9212%	-3.4351%
Vaccine rotation 2020	+0.8734%	+1.0169%
Brexit joint 2016	+0.9520%	+1.3543%
Dollar vortex 2022	+2.6950%	+3.6258%
Bear flattener 2022	+3.3440%	+3.9574%
Dollar squeeze 2020	+13.3444%	+15.1260%

Unit-equal Composite vs PCA2	+3.0479% - worse
Unit-equal Joint vs PCA2	+3.6076% - worse
Seeds with unit-equal improvement vs PCA2	0 / 5
Units improving vs PCA2 in all seeds	1 / 6

Conclusion The Reference copula was not a universal dependence solution. A structure that was excellent for one UST/rates-FX unit could be actively harmful for other units.

5. V5.1 – conservative 25% shrinkage

The next experiment kept the same rank-based idea but reduced its strength. Instead of replacing PCA2 dependence completely, the ordering score blended 75% native PCA2 ranks with 25% Reference ranks.

rank score = 0.75 x rank_PCA2 + 0.25 x rank_Reference

Development unit	Composite % vs PCA2	Joint % vs PCA2
Term-premium steepener 2023	-0.9431%	-1.1089%
Brexit joint 2016	-0.5770%	-0.8208%
Bear flattener 2022	-0.5480%	-0.6486%
Vaccine rotation 2020	-0.2724%	-0.3172%
Dollar vortex 2022	-0.0946%	-0.1273%
Dollar squeeze 2020	+0.8616%	+0.9767%

Unit-equal Composite vs PCA2	-0.2622%
Unit-equal Joint vs PCA2	-0.3410%
Seeds with unit-equal improvement	5 / 5
Units improving in every seed	5 / 6

Development read A weak shrinkage looked much safer than full replacement. At this point V5.1 was a credible candidate, but lambda=0.25 was frozen before fresh holdout validation.

6. Fresh eight-unit holdout – fixed shrinkage failed

The same 25% shrinkage was then applied to eight units that had not been used to choose the V5.1 setting.

Holdout unit	Composite % vs PCA2	Joint % vs PCA2
Inversion persistence 2019	-0.8690%	-1.6688%
Scandies stress 2022	-0.5668%	-0.7650%
Funding flip 2024	-0.2853%	-0.4056%
CPI shock cross 2022	+0.2225%	+0.3514%
Oil-FX split 2022	+0.3754%	+0.4746%
COVID curve 2020	+0.9601%	+1.1241%
Conundrum joint 2005	+0.9688%	+1.1606%
Safe-haven paradox 2011	+3.8963%	+7.3409%

Unit-equal Composite vs PCA2	+0.5877% - worse
Unit-equal Joint vs PCA2	+0.9515% - worse
Seeds with unit-equal Composite improvement	0 / 5
Seeds with unit-equal Joint improvement	0 / 5

Holdout conclusion The promising development result did not survive fresh units. Fixed shrinkage was therefore rejected rather than retuned on the holdout sample.

7. V5.2 – can historical performance choose the copula?

Since some units consistently benefited from shrinkage and others consistently did not, V5.2 tested a simple adaptive rule based only on a unit's own earlier walk-forward history.

First 12 folds: choose PCA2 or shrink25 -〉 freeze decision -〉 test on later 12 folds

Fresh unit	Selector decision	Later test result vs PCA2
Taper steepener 2013	PCA2	0.0000%
Hiking-year curve 2018	PCA2	0.0000%
Divergence 2014	shrink25	+5.8536%
Dots vs oil 2014	PCA2	0.0000%
BOJ-UST channel 2023	shrink25	-0.7186%
Rates into factors 2022	shrink25	+1.4473%
Fragile-five joint 2013	PCA2	0.0000%

Adaptive Composite vs PCA2	+0.9403%
Adaptive Joint vs PCA2	+1.3870%
Always-shrink25 Composite vs PCA2	+2.7958%
Units where adaptive beat PCA2	1 / 7
Selector directional accuracy	57.1%

8. Why the adaptive gate failed

The early-fold selector was not random, but it was not persistent enough. Several units reversed direction between the selector period and the later test period.

Example	Early selector evidence	Later outcome
Divergence 2014	shrink25 -1.51%	shrink25 +5.85% worse
Rates into factors 2022	shrink25 -0.92%	shrink25 +1.45% worse
Fragile-five 2013	selector chose PCA2	shrink25 later -1.36% better
BOJ-UST channel 2023	shrink25 -0.77%	shrink25 later -0.72% better

Generalization lesson Dependence preference was itself regime-dependent. Past relative performance of two copulas was not a reliable state variable for future relative performance.

At this stage, tuning the selector threshold or trying 10%, 20%, 30% shrinkage would turn the public-unit research set into a parameter-search target. The correct response was to stop the branch.

9. V5 final decision and V6 architecture

V5 was a valuable negative-result branch. It confirmed that Joint was a genuine modelling problem, but also showed that a generic Reference-style copula correction was not robust enough for production.

V5 experiment	Decision
V5.0 full Reference copula	Reject - strong single-unit result, poor cross-unit generalization
V5.1 fixed 25% shrinkage	Reject - development success, fresh holdout failure
V5.2 past-performance adaptive gate	Reject - selector accuracy only 57.1%, test Composite worse
PCA2 native dependence	Retain as the defensible default joint layer
V4.3 text-tail overlay	Retain as the conservative text-conditioned stress layer

V6 candidate = PCA2 native joint + V4.3 text-conditioned tail overlay

One-line conclusion

V5 showed what not to add. The final architecture should preserve the robust PCA2 dependence rather than forcing a copula correction that only works in selected regimes.

10. V5 research checklist – what was actually learned

The branch is best understood as a sequence of increasingly strict falsification tests rather than a sequence of failed parameter tweaks.

Question	Evidence-based answer
Was Joint a real bottleneck?	Yes. The development rank-reordering diagnostic changed Composite materially with zero marginal/tail change.
Is the Reference copula universally better?	No. Six-unit validation showed large and systematic failures.
Does weak shrinkage solve generalization?	No. 25% shrinkage failed on an eight-unit fresh holdout.
Can recent backtest performance select the right dependence rule?	Not reliably in the tested design; chronological selector accuracy was 57.1%.
Should V5 be discarded from the project story?	No. It is strong ablation evidence for retaining PCA2 native dependence.
Next step	Build V6 as an integration/finalization layer rather than another dependence search.

Research discipline The important decision was to stop tuning after fresh holdouts contradicted the development gains. This protects the final candidate from public-unit overfitting.